

Semisupervised graph U-Net with G-ConvLSTM for hyperspectral image classification

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ABSTRACT

Graph convolutional networks (GCNs) have demonstrated the effectiveness for hyperspectral image (HSI) classification, but still face challenges, such as insufficient exploitation of data structure information, limited labeled samples, and high susceptibility to noise and outliers. To address these issues, a semisupervised graph U-Net with graph convolutional long short-term memory is proposed for HSI classification, abbreviated as SSGU-Net. Specifically, we design a novel graph convolutional long short-term memory feature extractor to learn discriminative spatial-spectral joint features by simultaneously modeling the correlations in the spatial and spectral domains. Then, we develop a semisupervised graph U-Net with mutually inverse operation of the graph pooling and the graph unpooling modules which uses both labeled samples and unlabeled samples to train a well-parameterized network for HSI classification. In particular, to suppress the effects of noise and outliers, the graph pooling module is designed to selectively retain discriminative samples and fully learn the optimal correlation between these retained samples. Meanwhile, the graph unpooling module employs the local spatial context to reconstruct the reduced samples, thus restoring the pooled data to its original scale for classification task. Extensive experiments show the effectiveness of the proposed method, achieving overall accuracy gains of 5.22%, 1.58%, and 1.57% over the state-of-the-art competitors on the Indian Pines, University of Pavia, and Houston datasets, respectively.

1. Introduction

Hyperspectral images (HSIs) are characterized by hundreds of nearly continuous observations (spectral bands) in the electromagnetic spectrum from the visible to the near-infrared wavelength ranges (Barbin et al., 2012; Zhang et al., 2018). HSI classification is an essential task which assigns each pixel to a unique category by leveraging spatial and spectral information (Hu et al., 2021; Ma & Chang, 2022). Recently, HSI classification has been widely used in many practical applications, such as agriculture management (Gevaert et al., 2015; Sun et al., 2024), environmental protection (Li et al., 2024; Zhang et al., 2025), land monitoring (Tan et al., 2020), and other fields (Ding et al., 2025a; Kumar et al., 2024; Luo et al., 2022; Wang et al., 2025; Zhao et al., 2022b).

HSI classification is a challenging task due to the complex spatial distribution and the spectral heterogeneity of land covers. Traditional methods, such as support vector machine (SVM) (Martins et al., 2024),

random forest (RF) (Liu et al., 2017), sparse and collaborative representation (SCR) (Peng & Du, 2017), and multiple logistic regression (MLR) (Wang, 2022), rely on handcrafted feature extraction for HSI classification. However, these methods require expert knowledge (Ding et al., 2025a,b; Yang et al., 2024a). Different from these traditional methods, deep learning methods demonstrate superior capabilities in automatic feature extraction and model generalization (Zhong et al., 2020; Zhu et al., 2018). Among them, convolutional neural networks (CNNs), as one of the most classical methods, use trainable filters to extract local features in HSIs (Ghaderizadeh et al., 2021; Guo et al., 2020; Xu et al., 2024b). Specifically, as the first attempt, Chen et al. (2014) and Hu et al. (2015) converted the HSI into the 1-dimensional (1-D) sequence data, and then fed them into the 1-D convolution kernel to extract spectral features. Nevertheless, 1-D CNNs ignore the spatial information in HSIs. Therefore, Kiran Pandiri et al. (2024) and He et al. (2019) reduced the dimensionality of HSIs, processed pixels into 2-D windows, and

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introduced 2-D convolution kernel to learn the spatial features. These techniques, however, often fail to leverage the intrinsic 3-D structure information of HSI. Subsequently, some researchers cropped pixels into 3-D patches and utilized the 3-D convolutional kernel to effectively extract the integrated spectral-spatial features of HSIs (Bhatti et al., 2023; Li et al., 2022b; Liu et al., 2023; Yan et al., 2024a,b). In addition to CNNs, long short-term memory (LSTM) extracts spectral features by treating each pixel's spectral bands as a temporal sequence to capture long-range spectral dependencies in HSIs. Both CNN-based methods and LSTM-based methods exhibit satisfactory performance for HSI classification, but ignore the intrinsic geometric properties in HSIs.

To model geometric properties in HSIs, graph convolutional networks (GCNs) encode HSI into graph structure, and then extract the features from the relationships between land covers, thus achieving unprecedented classification performance (Wu et al., 2024; Xu et al., 2024a; Yang et al., 2022a). Qin et al. (2019) constructed spatial and spectral graphs from the relationships of pixels and fed them to GCN to extract spatial-spectral features. These pixel-level GCNs are able to capture the information from the graph of HSIs where pixels are treated as nodes and their relationships are used as edges (Hanachi et al., 2021; He et al., 2022; Hong et al., 2021; Mou et al., 2020). However, these methods require extensive storage memory and lead to huge computational costs, especially for large-scale remote sensing images. To address this issue, the superpixel-level GCN-based methods utilize simple linear iterative clustering (SLIC) to segment HSI and replace the original pixels with superpixel samples as nodes in the graph (Luo et al., 2024; Yang et al., 2024b; Yonghe et al., 2025; Zhen et al., 2025; Zhili et al., 2023). These two types of GCN-based methods have achieved promising results in HSI classification. However, these methods remain susceptible to over-smoothing, causing homogenized node features and compromised discriminative power across the graph (Shen et al., 2024; Yang et al., 2023).

To address the over-smoothing problem, graph U-Nets were further proposed to achieve deeper GCN while maintaining high-level feature extraction capabilities for classification (Gao & Ji, 2022; Gao et al., 2021). In this model, the graph pooling operation adaptively selects a subset of nodes to form a smaller graph to extract abstract feature and expand the receptive field. Moreover, the graph unpooling operation, as the inverse of graph pooling operation, restores the pooled graph to its original size by the position information of nodes. Subsequently, Guo et al. (2021) employed the graph pooling and unpooling operations in graph U-Nets to acquire effective spatial-spectral features. A multilevel superpixel structured graph U-Net was also developed to learn features at different scales from the spatial topology on multilevel graphs in HSIs (Liu et al., 2022). Zhao et al. (2022a) designed a U-shaped object graph neural network to extract hierarchical contextual features for multi-level image interpretation and HSI classification. Chen et al. (2025) proposed a graph U-Net with topology-feature awareness pooling, by simultaneously using topological structure and feature information to automatically select important nodes and extract compelling features for HSI classification. Zhang et al. (2024) developed a U-shaped semantic segmentation network that combines global spectral attention and deformable Transformer to extract multiscale global high-level features for segmenting large-scale HSIs. Zhu and Liu (2025) proposed a hierarchical graph U-Net with the global context awareness of the sparse transformer model, which innovatively integrates the local feature extraction capabilities for HSI classification. For ease of reading, Tables 1 and 2 provide the overview of the related methods and characteristics of the main related works, respectively.

Despite the promising results achieved by the aforementioned GCN-based methods, there are still some challenges to be solved. (1) HSIs contain rich spatial and spectral information, but GCN-based methods fail to fully utilize the correlation between different spectral bands in HSI data. (2) Acquiring labeled pixels in HSI requires expertise and is both time-consuming and costly. Thus, achieving high classification performance with a limited number of labeled samples is critically

important for practical applications. (3) Due to the interference of noise, the performance of GCN-based models may not be desirable.

In this paper, we propose a semisupervised graph U-Net with graph convolutional LSTM (G-ConvLSTM), abbreviated as SSGU-Net, for HSI classification. Specifically, to extract spatial-spectral features of HSIs and solve the above problem (1), we build a novel G-ConvLSTM as the feature extractor by integrating the graph convolution operation into the LSTM cell. Furthermore, to alleviate the above issue of limited labeled samples (2), a semisupervised graph U-Net that effectively uses both labeled and unlabeled samples is designed to train a well-parameterized network. In this semisupervised architecture, to suppress the effects of noise and outliers (i.e., changllge (3)), the graph pooling module retains high-quality samples by reducing some unlabeled interfering samples and explores the optimal relationship between the retained samples. The graph unpooling module learns the discriminative features of the reduced samples by spatial context information, thus restoring the pooled data to its original size for the classification task. Besides, skip connection is applied to mitigate the gradient vanishing problem. Experimental results show the effectiveness of the proposed method, achieving overall accuracies (OA) of 85.89%, 86.27%, and 76.26% with only five training samples per class for the Indian Pines, University of Pavia, and Houston datasets, respectively. In summary, the main contributions of this paper can be summarized as follows:

- To fully capture the joint spatial-spectral features of HSIs, we innovatively propose a G-ConvLSTM that integrates GCN for spatial feature extraction and LSTM for spectral sequence learning. To the best of our knowledge, the first novel integration of GCN and LSTM is used as a feature extractor to achieve discriminative spatial-spectral feature extraction for HSI classification.
- In view of the case with limited labeled samples, we propose a novel semisupervised graph U-Net framework that jointly uses a small number of labeled samples and a large number of unlabeled samples to significantly enhance data utilization efficiency. Consequently, a well-parameterized network is trained to improve the performance of HSI classification.
- In the semisupervised architecture, the graph pooling module is proposed to retain all labeled samples and discriminative unlabeled samples, while exploring the optimal relationship of the retained samples, to suppress the interference of noise. Meanwhile, our graph unpooling module employs the spatial context information rather than original skip connection to learn the features of the reduced samples, restoring the pooled data to its original size for classification.

The remainder of this paper is organized as follows. Section 2 introduces the relation work. Section 3 describes the proposed SSGU-Net method. Section 4 conducts a comprehensive quantitative analysis and evaluation of our method in comparison with other state-of-the-art approaches. Section 5 concludes the paper with some remarks.

2. Relation work

In this section, the reviews of LSTM-based methods and GCN-based methods for HSI classification are summarized as follows.

2.1. Long short-term memory-based hyperspectral image classification methods

Recurrent neural networks (RNNs) are proposed to efficiently process various time series data. However, in dealing with long sequences, RNNs suffer from gradient explosion and vanishing problems. LSTM, as a gated RNN, utilizes memory cells to model long-term dependencies in sequential data. Recently, LSTM has been gradually employed for HSI classification (Li et al., 2022a; Mou et al., 2017). In particular, Zhou et al. (2017) first attempted to apply LSTM for learning the spatial and spectral information from the spatial context and the spectral correlation of

Table 1
Overview on the related methods.

Method Category	Method	Advantages	Disadvantages
Traditional machine learning	SVM (Martins et al., 2024) RF (Liu et al., 2017) SCR (Peng & Du, 2017) MLR (Wang, 2022)	Computational efficiency; robustness to limited training samples; interpretable representations	Require specialized domain knowledge
	1-D CNN (Chen et al., 2014), (Hu et al., 2015)	Extract spectral features	
Deep learning	CNN 2-D CNN (Kiran Pandiri et al., 2024), (He et al., 2019) 3-D CNN (Bhatti et al., 2023), (Yan et al., 2024a), (Yan et al., 2024b), (Liu et al., 2023), (Li et al., 2022b)	Capture local spatial features Jointly extract spectral-spatial features by preserving 3D contextual relationships	Fail to employ the intrinsic geometric properties in HSI
	LSTM (Zhou et al., 2017), (Mei et al., 2022), (Xu et al., 2018), (Shi et al., 2015), (Hu et al., 2020), (Wang et al., 2022)	Effectively model long-range spectral dependencies in HSI	
	GCN Pixel-level GCN (Mou et al., 2020), (Hanachi et al., 2021), (He et al., 2022), (Hong et al., 2021) Superpixel-level GCN Zhili et al. (2023), (Yonghe et al., 2025), (Zhen et al., 2025), (Luo et al., 2024), (Yang et al., 2024b)	Capture features by modeling pixels as nodes and their correlations as edges Reduce storage requirements and computing time	Neglect the correlations between spectral bands in HSI
	Graph U-Net Gao and Ji (2022), (Gao et al., 2021), (Guo et al., 2021), (Liu et al., 2022), (Zhao et al., 2022a), (Chen et al., 2025), (Zhu & Liu, 2025)	Achieve deeper GCN architectures while preserving high-level feature extraction capabilities	

Table 2
Characterization of the main related works.

Method	Data	Feature extraction (Network)			Explore correlation	Training method	
	preprocessing	spatial	spectral	both	of spectral bands		
SSLSTM (Zhou et al., 2017)	PCA	✓			✓	Supervised	
GCN (Kipf & Welling, 2017)	—	LSTM				—	Semisupervised
		GCN					
MDGCN (Wan et al., 2020)	SLIC	✓				—	Semisupervised
DIGCN (Wan et al., 2022)		GCN					
FuNet (Hong et al., 2021)	—	✓			—	Supervised	
CEGCN (Liu et al., 2021)	SLIC	CNN; GCN			—	Semisupervised	
NESSGCN (Zhen et al., 2025)		CNN; GCN					
SSCL3DNN (Hu et al., 2020)	PCA	✓			✓	Supervised	
RSSGL (Wang et al., 2022)		CNN; LSTM					
MSSGU (Liu et al., 2022)	SLIC	✓			—	Semisupervised	
HGSTNet (Zhu & Liu, 2025)	SLIC, PCA	GCN; U-Net			—	Semisupervised	
		✓					
Proposed SSGU-Net	SLIC,	✓			✓	Semisupervised	
	band selection	LSTM; GCN; U-Net					

HSIs, respectively. A bidirectional LSTM-based network was proposed to explore both forward and backward spectral correlation within HSIs (Mei et al., 2022). To avoid the design of extremely deep LSTM networks, Xu et al. (2018) divided an HSI into several parts in the spectral direction and fed them into the LSTM. However, the above methods extract spatial and spectral features separately, resulting in ineffective

fusion of the two types of features. The combination of convolution and LSTM (ConvLSTM) has gradually become a solution to this problem (Shi et al., 2015). Hu et al. (2020) proposed a spatial-spectral ConvLSTM 2-D neural network (SSCL2DNN) for joint learning the spatial-spectral features and extended it to the corresponding 3-D version (SSCL3DNN) to better preserve the intrinsic structure of HSI data. Besides, a

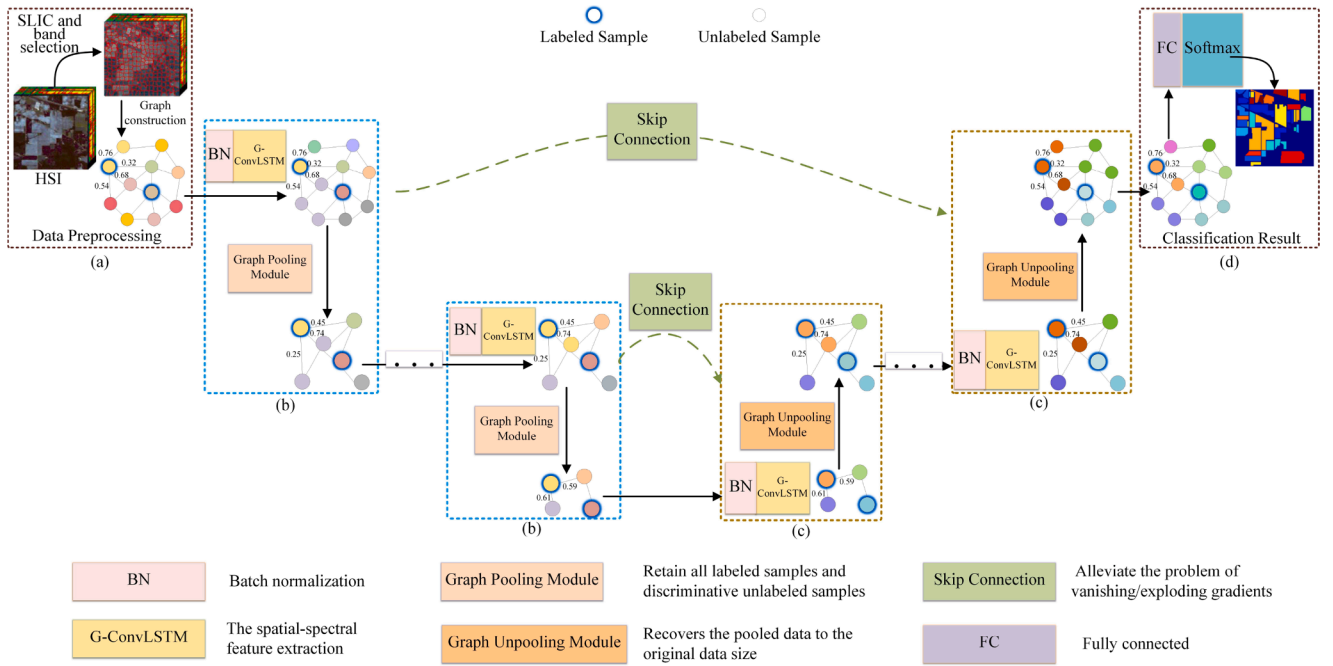


Fig. 1. The overall framework of the proposed SSGU-Net model. Data preprocessing is applied to the HSI to generate a superpixel graph in (a). It then fed into the layers in (b), each of which contains a BN, a G-ConvLSTM (see Fig. 2), and a graph pooling module (see Fig. 3) to retain discriminative samples. Next, (c) containing a BN layer, a G-ConvLSTM and a graph unpooling module (see Fig. 4), recovers the pooled data to the original data size. Finally, the learned features are fed into an FC layer and a Softmax in (d) to obtain the classification result.

regularized 3-D ConvLSTM global learning framework was developed to learn spectral correlation and global spatial context (Wang et al., 2022). However, these methods ignore the inherent topological information of HSIs.

2.2. Graph convolutional network-based hyperspectral image classification methods

Graph-structured data, as a representative of non-Euclidean spatial data, is widely used in a variety of fields such as social networks, e-commerce, knowledge graphs, chemistry, biology, computer vision, and linguistics, etc. Motivated by the benefits of deep learning for automated feature extraction, researchers are interested in developing a deep learning technique that can directly handles the graph-structured data. In 2013, Joan et al. (2014) proposed the earliest graph neural network based on spectral theory, defining graph convolution operations in spectral space. Based on this, Kipf and Welling (2017) proposed GCN by parameterizing the convolution kernel. The GCNs have revolutionized the paradigm of graph representation learning, enabling more effective analysis and understanding of graph structured data in various applications.

In recent years, GCNs effectively exploit the intrinsic geometric information in the field of HSI classification. For example, Qin et al. (2019) constructed spatial and spectral graphs from the relationship of pixels and fed them to the GCN to extract spatial-spectral features. By measuring the feature similarities between pixels, Mou et al. (2020) designed a nonlocal graph to learn the nonlocal representation of HSIs. Besides, Hanachi et al. (2021) combined deep autoencoder model and GCN for automatically extracting relevant features via deeper layers. He et al. (2022) utilized one GCN to fully extract features in HSI samples and another to implement label distribution learning, thus reducing the required number of training samples. In these methods, each pixel is regarded as a graph node and graphs are constructed over all the pixels of the HSI, leading to a high computational cost. As such, Hong et al. (2021) performed a minibatch strategy for GCN in HSI clas-

sification. Wan et al. (2020) segmented the HSIs into a small amount of compact superpixels and treated each superpixel as a graph node. Subsequently, Wan et al. (2022) introduced the dual GCN branches to explore the contextual information at different spatial scales and work interactively to refine the graph information. Liu et al. (2021) proposed a CNN-enhanced GCN model where CNN and GCN branches respectively learn features from small-scale regular regions and large-scale irregular regions, and generate complementary spectral-spatial features, respectively. In addition, a new GCN spatial-spectral feature extraction technique implemented graph convolution on every local region of HSIs (Zhang et al., 2022). These methods ignore an adaptive and automated way of choosing the receptive fields or paths of GCN. To guide the receptive paths, Liu and other (2019) proposed an adaptive path layer with adaptive breadth and depth functions. However, these GCN-based HSI classification methods ignore the correlation between the spectral bands in HSIs.

In pursuit of overcoming the issue of insufficient exploitation of data structure information in HSIs, we propose a novel SSGU-Net for HSI classification. Specifically, to learn the joint spatial and spectral features, a G-ConvLSTM is designed by simultaneously modeling the spatial topological information and the correlation between spectral bands in HSIs. Besides, a semisupervised graph U-Net uses small labeled data and massive unlabeled data to train a well-parameterized network for HSI classification.

3. Proposed method

The overall framework of the proposed SSGU-Net model is shown in Fig. 1. Specifically, Section 3.1 introduces data preprocessing. G-ConvLSTM is built in Section 3.2. In Section 3.3, a semisupervised graph U-Net including graph pooling module, graph unpooling module, and skip connection is presented in detail. The loss function is provided in Section 3.4. Finally, Section 3.5 describes the overall framework of the proposed SSGU-Net model. We summarize the important abbreviations in this paper via Table 3.

3.1. Data preprocessing

Given an original HSI $X_o \in \mathbb{R}^{N_o \times M_o}$, N_o and M_o are the numbers of pixels and spectral bands. The large number of pixels in HSI data and the presence of redundant spectral bands may result in high computational complexity. To address this, we apply the simple linear iterative clustering (SLIC) method (Wan et al., 2020) and band selection method (Yin et al., 2012) to the original HSIs, respectively. The processed data is represented as $X = \{x_1, \dots, x_i, \dots, x_N\} \in \mathbb{R}^{N \times M}$. Here, the feature vector x_i corresponds to the i th superpixel and is defined by the average of its all pixels. N denotes the number of superpixels. M represents the number of selected spectral bands. As shown in the Table 13, these preprocessing methods are effective in reducing the runtime and improving the HSI classification performance.

In GCN-based HSI classification methods, transforming the HSI into graph structured data is a critical prerequisite for effective feature extraction. After applying SLIC and band selection, each superpixel is regarded as a node in the constructed graph. Consequently, X serves as the feature matrix (Table 4). The adjacency matrix A indicates relationships between superpixels, which is calculated by

$$A_{ij} = \begin{cases} e^{-\gamma \|x_i - x_j\|^2}, & \text{if } x_i \in \mathcal{N}(x_j) \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where A_{ij} represents the weight between the i th sample x_i and the j th sample x_j . The parameter γ is empirically set to 0.2 as suggested in (Wan et al., 2020). $\mathcal{N}(x_j)$ is the set of neighbors of x_j .

3.2. G-ConvLSTM

HSI is a 3-D spectral-spatial data, which integrates the spectral information with the 2-D spatial information of land covers. The spatial domain denotes the location coordinates of the pixels. The spectral domain represents the spectral features of different bands. In the paper, a novel G-ConvLSTM is designed as the feature extractor to extract the fused spatial-spectral features from the spatial geometry properties of pixels and the spectral correlations of different bands. The structure of the G-ConvLSTM in the l th layer is illustrated in Fig. 2.

Let $X^l = [X_1^l, X_2^l, \dots, X_M^l]$ denote the spectral matrix of HSI in the l th layer, where M is the number of bands. First of all, a grouping strategy (Xu et al., 2018) decomposes the features of HSI in the l th layer into T components with the same dimension $m = \text{floor}(M/T)$ along the spectral axis, yielding grouped subsets $\{H_1^l, \dots, H_t^l, \dots, H_T^l\}$, $t \in \{1, 2, \dots, T\}$.

Table 3
List of abbreviations.

Abbreviation	Explanation
GCN	Graph convolutional network
HSI	Hyperspectral image
LSTM	Long short-term memory
SVM	Support vector machine
SCR	Sparse and collaborative representation
MLR	Multiple logistic regression
RF	Random forest
CNN	Convolutional neural network
1-D	1-Dimensional
PCA	Principal component analysis
G-ConvLSTM	Graph convolutional long short-term memory
SSGU-Net	Semisupervised graph U-Net with G-ConvLSTM
RNN	Recurrent neural network
ConvLSTM	Convolution and LSTM
SLIC	Simple linear iterative clustering
BN	Batch normalization
FC	Fully connected
OA	Overall accuracy
AA	Average accuracy
κ	Kappa coefficient

Table 4

Mathematical notations used in the proposed method.

Symbols	Description
X_o	Feature matrix for the original HSI
N_o	Number of pixels
M_o	Number of spectral bands
X	Feature matrix for the preprocessed HSI
N	Number of superpixels
M	Number of selected spectral bands
A	Adjacency matrix for superpixel graph
A^l	Adjacency matrix in the l th layer
$\tilde{A}^l$	Self-connected adjacency matrix in the l th layer
D^l	Diagonal degree matrix of $\tilde{A}^l$
T	Number of cells in the G-ConvLSTM
m	Number of spectral bands of each component in the G-ConvLSTM
H_t^l	t th continuous band feature
Z_t^l	Spatial features extracted by GCN in the t th G-ConvLSTM cell in the l th layer
H_{out}^l	Output of the t th G-ConvLSTM cell in the l th layer
H_{out}^l	Spatial-spectral features are learned by the G-ConvLSTM in the l th layer
$Z_{labeled}^l$	Binary mask matrix for the labeled samples in the l th layer
$H_{labeled}^l$	Feature matrix where the features of unlabeled samples are set to 0 in the l th layer
$id_{labeled}^{x_l}$	Indices of labeled samples in the l th layer
$Z_{unlabeled}^l$	Binary mask matrix for the unlabeled samples in the l th layer
$H_{unlabeled}^l$	Feature matrix where the features of labeled samples are set to 0 in the l th layer
$id_{unlabeled}^{x_l}$	Indices of the unlabeled samples in the l th layer
$id_{retained}^{x_l}$	Indices of retained samples in the l th layer
$id_{reduced}^{x_l}$	Indices of reduced samples in the l th layer
A_{in}^{l+1}	Adjacency matrix of the retained samples in the $(l+1)$ th layer
H_{empty}^{l+1}	Empty feature matrix in the $(l+1)$ th layer
$H_{retained}^{l+1}$	Feature matrix where the feature of reduced samples are 0 in the $(l+1)$ th layer
Q^l	Spatial relationship between the reduced samples and the selected samples in the l th layer

This prevents excessive network width from the traditional band-by-band strategy.

$$H_t^l = [X_{(t-1)m+1}^l, X_{(t-1)m+2}^l, \dots, X_{tm}^l] \quad (2)$$

where H_t^l denotes the t th component of the features of HSI in the l th layer, which contains m continuous spectral bands. $X_{(t-1)m+1}^l$ represents the $((t-1)m+1)$ th spectral band.

Then, these decomposed data are fed into the cells of the proposed G-ConvLSTM one by one. Each G-ConvLSTM cell has a similar calculational model that of LSTM, with significant innovation in introducing graph convolution operation in each gate unit. This integration of GCN and LSTM enables the model to effectively capture spatial-spectral features. For example, in the t th cell, the input gate i_t^l , the forget gate f_t^l , the output gate o_t^l , the input cell state $\tilde{c}_t^l$, the final cell state c_t^l , and the output of the t th G-ConvLSTM cell $H_{out_t}^l$ are updated by

$$\begin{aligned} Z_t^l &= D^{l-\frac{1}{2}} \tilde{A}^l D^{l-\frac{1}{2}} H_t^l \\ i_t^l &= \sigma(Z_t^l W_{ii}^l + H_{out_{t-1}}^l W_{(t-1)i}^l + b_i^l) \\ f_t^l &= \sigma(Z_t^l W_{if}^l + H_{out_{t-1}}^l W_{(t-1)f}^l + b_f^l) \\ o_t^l &= \sigma(Z_t^l W_{io}^l + H_{out_{t-1}}^l W_{(t-1)o}^l + b_o^l) \\ \tilde{c}_t^l &= \tanh(Z_t^l W_{ic}^l + H_{out_{t-1}}^l W_{(t-1)c}^l + b_c^l) \\ c_t^l &= f_t^l \odot c_{t-1}^l + i_t^l \odot \tilde{c}_t^l \\ H_{out_t}^l &= o_t^l \odot \tanh(c_t^l) \end{aligned} \quad (3)$$

where $\tilde{A}^l = A^l + I$ is the l th layer adjacency matrix with self-loop. D^l is the diagonal degree matrix of $\tilde{A}^l$. Z_t^l represents the learned spatial features by the graph convolution operation. $\sigma(\cdot)$ denotes the sigmoid function. $\tanh(\cdot)$ is the hyperbolic tangent function. $\odot$ denotes the element-wise multiplication. The weight matrices W_{ii}^l , W_{if}^l , W_{io}^l , and W_{ic}^l map the extracted spatial information to the corresponding gate. $H_{out_{t-1}}^l$ is

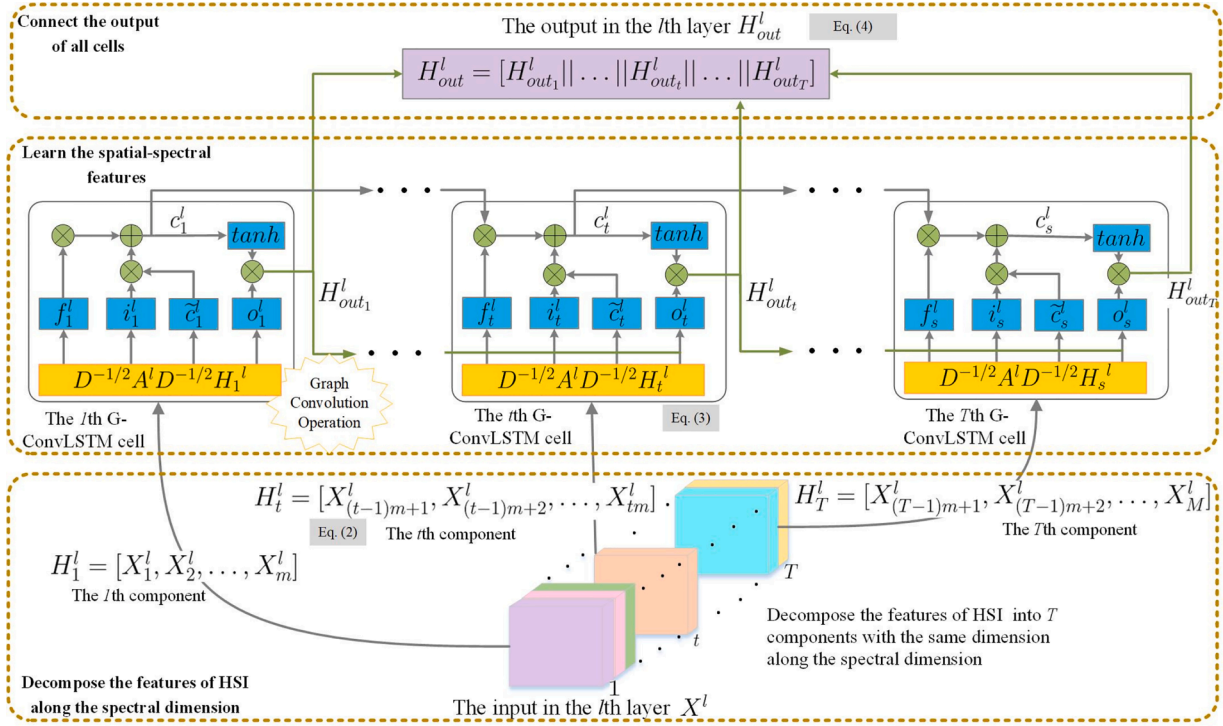


Fig. 2. Illustration of the G-ConvLSTM. Specifically, the features of HSI in the \$l\$th layer are decomposed into \$T\$ components along the spectral dimension. These decomposed components are then sequentially input into each cell of the G-ConvLSTM to learn spatial-spectral features. Finally, concatenating the outputs of all G-ConvLSTM cells yields the final output.

the output of the \$(l-1)\$th cell. \$W_{(t-1)i}^l\$, \$W_{(t-1)f}^l\$, \$W_{(t-1)o}^l\$, and \$W_{(t-1)c}^l\$ are the weight matrices for the preceding hidden state. The bias terms are represented by \$b_i\$, \$b_f\$, \$b_o\$, and \$b_c\$, respectively.

The above G-ConvLSTM cell serves as the fundamental unit to build the proposed G-ConvLSTM. Therefore, the output of the G-ConvLSTM in the \$l\$th layer is defined as

$$H_{out}^l = [H_{out_1}^l || \dots || H_{out_t}^l || \dots || H_{out_T}^l] \quad (4)$$

where \$H_{out}^l\$ denotes the output of the proposed G-ConvLSTM in the \$l\$th layer, which is learned via concatenation of the outputs of all G-ConvLSTM cells in the \$l\$th layer. \$||\$ represents cascading operation.

Fig. 2 depicts the architecture and dataflow of the G-ConvLSTM module. Specifically, the features \$X^l\$ of HSI in the \$l\$th layer are decomposed into \$T\$ components along the spectral dimension, denoted as \$\{H_1^l, \dots, H_t^l, \dots, H_T^l\}\$. Subsequently, these components are fed one by one into each cell of the G-ConvLSTM to learn spatial-spectral features. Finally, concatenating the outputs \$H_{out_1}^l, \dots, H_{out_t}^l, \dots, H_{out_T}^l\$ of all G-ConvLSTM cells in the \$l\$th layer yields the final output \$H_{out}^l\$.

3.3. Semisupervised graph U-Net

The performance of graph U-Net is usually unsatisfactory or even poor when trained on only limited number of labeled samples (Gao & Ji, 2022). To address the issue, we propose a semisupervised graph U-Net by leveraging both labeled and unlabeled samples to build better classifier and improve the performance for HSI classification. The semisupervised architecture contains graph pooling module, graph unpooling module, and skip connection.

1) Graph Pooling Module: As illustrated in Fig. 3, a novel graph pooling module retains all labeled samples and discriminative unlabeled samples in semisupervised architecture by filtering out unlabeled interfering samples (e.g., noise, outliers). Besides, the module also explores

the optimal relationships among retained samples to capture more effective spatial-spectral information.

In the semisupervised graph U-Net architecture, both labeled and unlabeled samples are utilized for network training. Labeled samples serve as the foundational supervised signal in semisupervised graph-based learning, enabling the model to establish feature representations and propagate discriminative information through the graph structure. Therefore, to retain labeled samples in network, our graph pooling module introduces a binary mask matrix \$Z_{labeled}^l\$, where each element \$z_{ij} \in \{0, 1\}\$ corresponds to the label status of the sample. As a result, the indices of labeled samples are computed by

$$\begin{aligned} H_{labeled}^l &= H_{out}^l \odot Z_{labeled}^l \\ idx_{labeled}^l &= F_{find}(H_{labeled}^l) \end{aligned} \quad (5)$$

where \$H_{out}^l\$ represents the spatial-spectral feature matrix learned by Eq. (4). \$\odot\$ denotes the element-wise multiplication. \$H_{labeled}^l\$ is the feature matrix in which the features of unlabeled samples are set to zero. The operator \$F_{find}(\cdot)\$ finds rows with non-zero values in \$H_{labeled}^l\$, yielding indices \$idx_{labeled}^l\$ of labeled samples.

Additionally, for unlabeled samples, the goal of the graph pooling module is to selectively retain those exhibiting discriminative information. This strategy mitigates interference from low-information samples, thereby enabling the model to more effectively learn features and enhance classification accuracy. To achieve this, we utilize a trainable projection vector to map all node features to 1D. Unlabeled samples with high projection scores are identified as discriminative and subsequently propagated to the next layer for more efficient spatial-spectral feature extraction. Specifically, a binary mask \$Z_{unlabeled}^l\$ is constructed to isolate unlabeled samples. A feature matrix \$H_{unlabeled}^l\$ is generated by masking labeled sample features to zero. A trainable projection vector is employed to compute retention scores. Samples with higher retention scores in vector \$y_{unlabeled}^l\$ are propagated to the next network layer. The

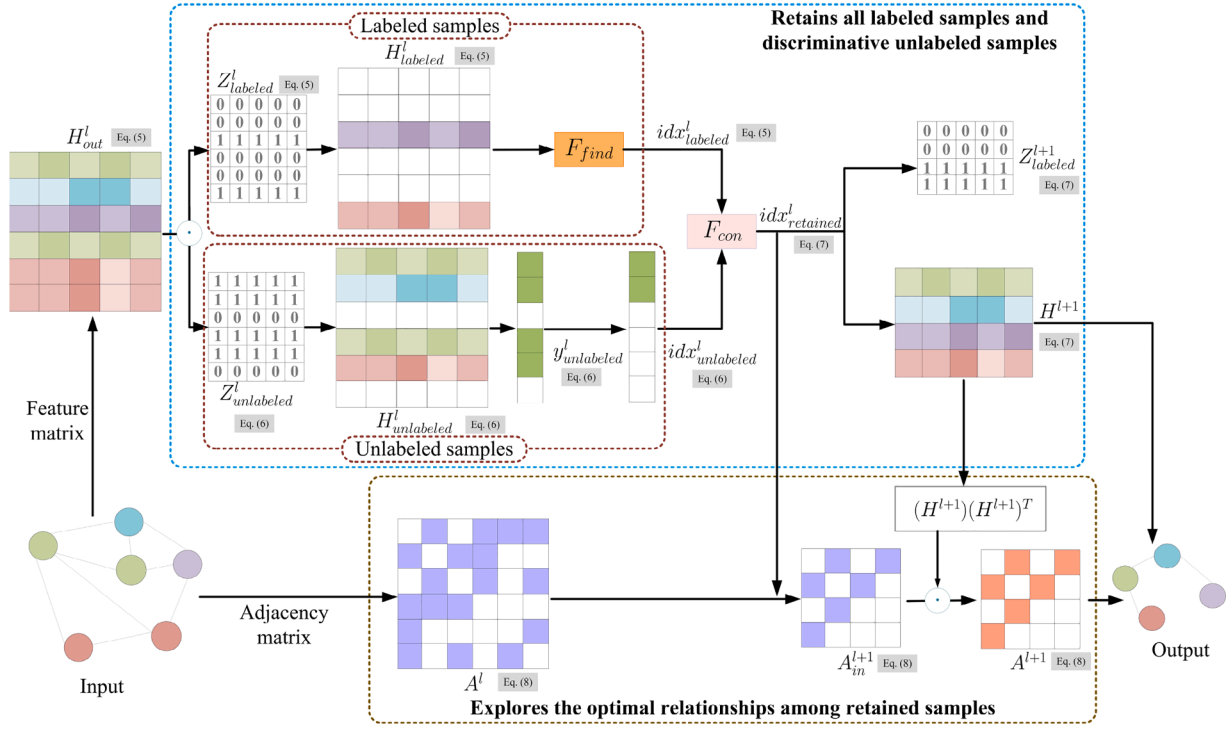


Fig. 3. Structure of the graph pooling module. The graph pooling module retains all labeled samples and discriminative unlabeled samples, while also exploring the optimal relationships among retained samples.

indices of the retained unlabeled samples are computed as

$$\begin{aligned} H_{unlabeled}^l &= H_{out}^l \odot Z_{unlabeled}^l \\ y_{unlabeled}^l &= H_{unlabeled}^l p^l \\ idx_{unlabeled}^l &= \text{rank}(y_{unlabeled}^l, kN^l - N_{labeled}) \end{aligned} \quad (6)$$

where H_{out}^l represents the spatial-spectral feature matrix learned by Eq. (4). $Z_{unlabeled}^l$ is a binary mask matrix, which is obtained by $Z_{unlabeled}^l = 1 - Z_{labeled}^l$. $\odot$ denotes the element-wise multiplication. $H_{unlabeled}^l$ is the feature matrix in which the features of labeled samples are set to zero. $y_{unlabeled}^l$ denotes the retention score vector. p^l is a trainable parameter in the l th layer. $\text{rank}(y_{unlabeled}^l, kN^l - N_{labeled})$ returns the indices $idx_{unlabeled}^l$ of the top $kN^l - N_{labeled}$ values of the unlabeled samples in $y_{unlabeled}^l$. k is the proportion of retained samples among all samples. N^l is the number of all samples in the l th layer. $N_{labeled}$ is the total number of labeled samples.

Through the operations discussed above, the indices $idx_{labeled}^l$ of labeled samples and the indices $idx_{unlabeled}^l$ of retained unlabeled samples are obtained. The indices of all retained samples are obtained by connecting the above two indices. Thus, the feature matrix H^{l+1} and the binary mask matrix $Z_{labeled}^{l+1}$ in the $(l+1)$ th layer can be formulated as

$$\begin{aligned} idx_{retained}^l &= F_{con}(idx_{labeled}^l, idx_{unlabeled}^l) \\ H^{l+1} &= H_{out}^l(idx_{retained}^l, :) \\ Z_{labeled}^{l+1} &= Z_{labeled}^l(idx_{retained}^l, :) \end{aligned} \quad (7)$$

where the operator $F_{con}(\cdot)$ concatenates $idx_{labeled}^l$ and $idx_{unlabeled}^l$ to generate the indices $idx_{retained}^l$ of retained samples. $H_{out}^l(idx_{retained}^l, :)$ and $Z_{labeled}^l(idx_{retained}^l, :)$ perform row extraction to learn the feature matrix H^{l+1} and the binary mask matrix $Z_{labeled}^{l+1}$.

Acquiring an optimal relationship between all retained samples is another important task in our method. Due to the effects of the noise and complex scenes on HSI data, the graph in each graph pooling layer derived from the initial graph constructed in Eq. (1) is inaccurate. To obtain the optimal graph, we adaptively construct a new graph by lever-

aging the feature similarity, then fusing it with the graph from the initial graph. The corresponding adjacency matrix in the $(l+1)$ th layer can be computed by

$$\begin{aligned} A_{in}^{l+1} &= A^l(idx_{retained}^l, idx_{retained}^l) \\ A^{l+1} &= (H^{l+1})(H^{l+1})^T \odot A_{in}^{l+1} \end{aligned} \quad (8)$$

where A_{in}^{l+1} is the adjacency matrix of the retained samples, obtained by performing row and column extractions from the adjacent matrix A^l of the l th layer. The adjacency matrix of the new graph is constructed by using the extracted feature matrix H^{l+1} and its transposition $(H^{l+1})^T$. The adjacency matrix A^{l+1} of the optimal graph enables the G-ConvLSTM to extract more effective spatial-spectral features for HSI classification.

The process of the graph pooling module is shown in Fig. 3. Specifically, the input data contain the feature matrix H_{out}^l and the adjacency matrix A^l in the l th layer. In the first branch, the indices $idx_{labeled}^l$ of labeled samples are learned by the operation F_{find} . Concurrently, the indices $idx_{unlabeled}^l$ of the retained unlabeled samples are obtained via a learnable projection vector. F_{con} concatenates these indices to obtain the indices $idx_{retained}^l$ of retained samples, and then the feature matrix H^{l+1} of retained samples is learned by row-wise selection. In the second branch, the optimal graph A^{l+1} is obtained by fusing the graph A_{in}^{l+1} derived from A^l with a new graph constructed via feature similarity $(H^{l+1})(H^{l+1})^T$.

2) Graph Unpooling Module: The goal of graph pooling module is to select a subset of nodes in the graph. Conversely, the graph unpooling operation, as the inverse of the graph pooling operation, restores the pooled data to its original size for classification task. To achieve it, our graph unpooling module, shown in Fig. 4, exploits the spatial context information to learn the features of the reduced samples in the corresponding graph pooling module. Then, based on the positional indices of the retained samples and the indices of the reduced samples stored in the corresponding graph pooling module, we restore the pooled data to its original scale for classification task.

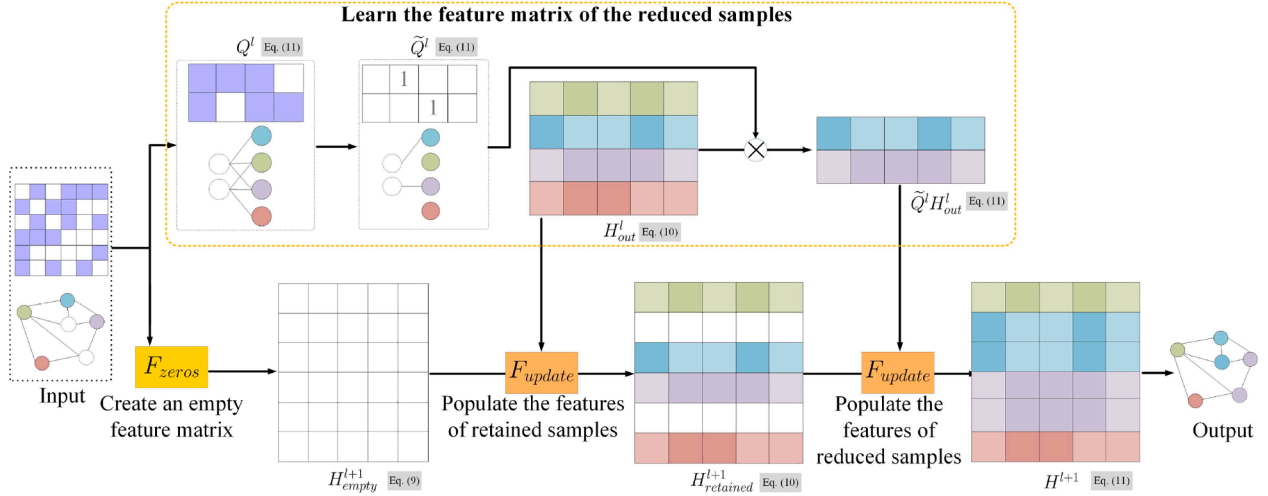


Fig. 4. Illustration of the graph unpooling module. $\otimes$ is the matrix multiplication operation. The graph unpooling module leverages spatial context information to effectively learn the features of reduced samples. By utilizing the positional indices of both the retained and reduced samples, it reconstructs the pooled data to its original size.

Firstly, an empty feature matrix is created as

$$H_{empty}^{l+1} = F_{zeros}(N^{l+1}, M) \quad (9)$$

where N^{l+1} is the number of samples in the corresponding graph pooling module, and M represents the feature dimension. The function F_{zeros} generates an initially empty matrix $H_{empty}^{l+1} \in \mathbb{R}^{N^{l+1} \times M}$.

Then, the graph unpooling module utilizes the indices stored in the corresponding graph pooling module to place the features of retained samples into the empty feature matrix. This operation can be denoted as

$$H_{retained}^{l+1} = F_{update}(H_{empty}^{l+1}, idx_{retained}^l, H_{out}^l) \quad (10)$$

where $idx_{retained}^l$ represents the indices of the retained samples in the corresponding graph pooling module. H_{out}^l denotes the feature matrix of the retained samples, learned via Eq. (4). The operation $F_{update}(\cdot)$ populates row vectors in H_{out}^l into H_{empty}^{l+1} according to their corresponding indices stored in $idx_{retained}^l$. In $H_{retained}^{l+1}$, row vectors with indices in $idx_{retained}^l$ are updated by row vectors in H_{out}^l , while other row vectors remain zero.

In addition to the features of the retained samples discussed above, the task of learning the features of the reduced samples is also an important issue that requires further investigation. In graph U-Net method (Guo et al., 2021), the graph unpooling operation utilizes skip connection to learn the features of the reduced samples based on information from the corresponding pooling module. However, these features exhibit weakly discriminative power for HSI classification. To tackle this, a novel strategy is proposed to obtain the features of the reduced samples from the retained samples with discriminative features by using the spatial similarity between samples. Specifically, two pixels closer in spatial distribution tend to have similar characteristics (Tobler's first law of geography) and belong to the same category. Based on this, the features of the reduced samples are learned from retained samples in the nearest neighborhood of the local neighborhood. Subsequently, the features of the reduced samples are placed in the feature matrix $H_{retained}^{l+1}$, according to

$$\begin{aligned} Q^l &= A^l(idx_{reduced}^l, idx_{retained}^l) \\ \tilde{Q}_{i,:}^l &= \begin{cases} 1 & \text{large}(Q_{i,:}^l, 1) \\ 0 & \text{otherwise} \end{cases} \\ H^{l+1} &= F_{update}(H_{retained}^{l+1}, idx_{reduced}^l, \tilde{Q}^l H_{out}^l) \end{aligned} \quad (11)$$

where A^l denotes the adjacency matrix in the previous layer of the corresponding graph pooling module. $idx_{reduced}^l$ and $idx_{retained}^l$ represent the indices of the reduced and retained samples, respectively, in the corresponding graph pooling module. Q^l is the relationship between the reduced samples and the retained samples, where each row represents the similarity between a reduced sample and all retained samples. $\text{large}(\cdot)$ is defined for setting the maximum value of each row in Q^l to 1 and other values to 0, learning a new matrix $\tilde{Q}^l$. H_{out}^l denotes the feature matrix of the retained samples, learned via Eq. (4). The learned feature matrix for the reduced samples is obtained by the matrix multiplication operation of $\tilde{Q}^l$ and H_{out}^l . F_{update} populates $\tilde{Q}^l H_{out}^l$ into $H_{retained}^{l+1}$ via the indices $idx_{reduced}^l$ of reduced samples to yield H^{l+1} .

The graph unpooling module is visualized in Fig. 4. Specifically, an empty matrix H_{empty}^{l+1} is initialized by F_{zeros} . Next, the feature matrix H_{out}^l of retained samples is placed into this empty matrix via F_{update} to generate $H_{retained}^{l+1}$. The features of reduced samples are obtained from these retained samples through neighborhood information aggregation $\tilde{Q}^l H_{out}^l$, and embedded into $H_{retained}^{l+1}$ to yield the final feature matrix H^{l+1} .

3) Skip Connection: Stacking multiple layers in the GCN-based methods typically causes the problem of vanishing/exploding gradients. To alleviate this problem, skip connection can be applied between layers. In the proposed SSGU-Net, skip connection connects the graph pooling module and corresponding graph unpooling module for preserving the low-level semantic information.

3.4. Loss function

Finally, the output classification results of SSGU-Net can be learned by $O = \text{Softmax}(H^{l+1}W^{l+1})$, where H^{l+1} is learned by Eq. (11) and W^{l+1} is the trained matrix. To train this network, the commonly used cross-entropy loss function is applied to reduce the difference between the network predicted labels and the ground truth of the training samples. Hence, the objective function of SSGU-Net can be formulated as

$$Loss = - \sum_{f=1}^C Y_f \log(O_f) \quad (12)$$

where C denotes the number of training samples. Y_f denotes the true label of the f th training sample. O_f denotes the predicted label of the f th training sample.

Algorithm 1 The proposed SSGU-Net method for HSI classification.

Input: HSI data: X_0 , Ground truth: Y , number of iterations: τ , learning rate η , the number of spectral bands of each component in the G-ConvLSTM: m , the proportion of selected samples from the whole set of samples in the graph pooling module: k , and the number of the network layers: L ;

- 1: Filter out redundant bands by band selection and segment the whole image into superpixels via SLIC algorithm to learn the feature matrix X ;
- 2: Construct the adjacency matrix A via Eq. (1);
- 3: Parameter setting and weights initialization;
- 4: **for** epoch $\leq \tau$ **do**
- 5: Calculate the feature matrix H^1 and the adjacency matrix A^1 through a sequential processing pipeline comprising: batch normalization, G-ConvLSTM as defined in Eqs. (2), (3), and (4), and graph pooling module according to Eqs. (5), (6), (7), and (8);
-
- 6: Calculate the feature matrix H^L via batch normalization, G-ConvLSTM (Eqs. (2), (3), and (4), and graph unpooling module (Eqs. (9), (10), and (11));
- 7: Calculate the network output O by a fully connected layer and softmax;
- 8: Obtain the value of the objective function in Eq. (12) and update the network parameters using full-batch gradient descent;
- 9: **end for**
- 10: Conduct label prediction based on the trained network;

Output: Predicted label for each pixel.

3.5. SSGU-Net framework

Graph U-Net has been successfully applied for HSI classification, but still faces challenges such as limited performance with small training sets and underutilization of spectral correlation. The SSGU-Net model is proposed to solve these limitations. As shown in Fig. 1, our method uses the proposed semisupervised graph U-Net as the base framework and the G-ConvLSTM as the feature extractor. The implementation detail of our SSGU-Net is shown in Algorithm 1.

In Algorithm 1, steps 1 and 2 are data preprocessing. During this phase, we apply SLIC and band selection on the original HSI data to obtain the feature matrix and adjacency matrix. Steps 3 to 9 are the model training process. In these steps, the cross-entropy loss function aims to achieve automatic learning of model parameters by minimizing the difference between the prediction labels generated by the network and the true labels of the training samples. Finally, step 10 obtains the labels for the test data using the trained model.

4. Experimental results

In the Section, exhaustive experiments are conducted on three public HSI datasets to validate the performance of the proposed SSGU-Net method. Some state-of-the-art methods are used for comparison, where the per-class accuracy, OA, average accuracy (AA), and Kappa coefficient (κ) are utilized as quantitative metrics to evaluate the classification performance of all methods.

4.1. HSI datasets

Three real HSI datasets, namely, Indian Pines, University of Pavia, and Houston, are chosen to evaluate the classification performance of our method. Their detailed descriptions are provided below.

1) *Indian Pines*: The dataset was recorded by the Airborne Visible-Infrared Imaging Spectrometer (AVIRIS) over the Indian Pines test site in North-western Indiana in 1992. It contains 145×145 pixels with a spatial resolution of 20 m per pixel and 220 spectral bands in the wave-

length ranging from 0.4 to 2.5 μm . 200 bands are used for all experiments, with 20 water absorption and noisy bands removed. This image is mainly used for agricultural related studies, and contains two-third of agricultural lands, one-third of forests, and other natural perennial vegetation, for a total of 16 land-cover classes.

2) *University of Pavia*: This dataset was captured by the Reflective Optics Spectrographic Imaging System (ROSIS-3) sensor over the area of the Pavia University in Italy, in 2001. The image is made up of 610×340 pixels with high spatial resolution of 1.3 m per pixel. After removing the noisy spectral bands, 103 bands are preserved with a spectral coverage ranging from 0.43 to 0.86 μm . There are 9 classes of urban land covers.

3) *Houston*: The dataset was gathered with the Compact Airborne Spectrographic Imager (CASI) sensor over the University of Houston, TX, USA, and the surrounding area on June 2012. It was introduced in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion contest. Houston data have a spatial size of 349×1905 pixels with a spatial resolution of 2.5 m, and 144 spectral bands with the wavelength ranging from 0.38 to 1.05 μm . There exist 15 land cover classes in this image.

To test the classification performance of the proposed method under limited labeled samples, five samples per class are randomly selected as training samples for each HSI dataset. Meanwhile, the rest of the samples in each class are regarded as the testing set to evaluate the performance of all methods. Fig. 5 shows the false-color maps, the corresponding ground-truth maps, the number of training pixels, and the total number of pixels for the Indian Pines, University of Pavia, and Houston datasets, respectively.

4.2. Compared methods

To evaluate the classification ability of our SSGU-Net, other recent state-of-the-art HSI classification methods are utilized for comparison. Specifically, five GCN-based methods, such as GCN (Kipf & Welling, 2017), MDGCN (Wan et al., 2020), DIGCN (Wan et al., 2022), ACGCN (Yang et al., 2022b), and MSSGU (Liu et al., 2022) are selected as comparative methods. Meanwhile, we also compare the proposed method with two HSI classification methods that combine the CNN and GCN such as FuNet (Hong et al., 2021) and CEGCN (Liu et al., 2021). The parameter settings for compared methods are described below.

1) GCN (Kipf & Welling, 2017): a graph convolutional hidden layer with 128 units is utilized before feeding the features into the softmax layer, where the adjacency matrix A is computed by using a K-nearest neighbors (KNN) with $K = 10$.

2) FuNet (Hong et al., 2021): the initially learning rate is set as 0.001 by $(1 - (\text{iter}/\text{maxiter}))^{0.5}$ at intervals of 50 epochs. In the process of network training, the maximum number of epochs is set to 200, batch normalization is adopted with the 0.9 momentum, and the batch size in the training phase is set to 32. Moreover, the l2-norm regularization, set to 0.001, is employed on weights to stabilize the network training and reduce overfitting.

3) MDGCN (Wan et al., 2020): three neighborhood scales are adopted, in each of which, two graph convolution layers are used with 20 hidden units. The learning rate is set to 0.0005 and the number of training epochs is set to 5000.

4) DIGCN (Wan et al., 2022): the number of iterations is fixed at 500, the learning rate is set to 0.001, the number of hidden units is 80, and the neighborhood sizes s_1 and s_2 are set to 1 and 2.

5) ACGCN (Yang et al., 2022b): the spectral dimension after PCA is set to 20. To mitigate overfitting, a dropout rate of 0.5 is applied to each graph convolutional layer. For Sa-GCN and Se-GCN, the dimensions of the two graph convolutional layers are set as 40 and 20, respectively. The epoch is set to 3000. The learning rate of 0.005 is used.

6) CEGCN (Liu et al., 2021): the segmentation scale λ is fixed to 100. The network depth of CEGCN is set to 2. The special size of convolutional kernels in the PCsN is fixed to 5×5 . All the activation functions are the

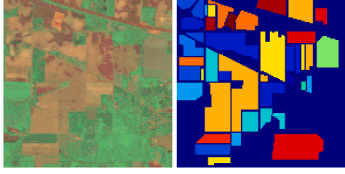
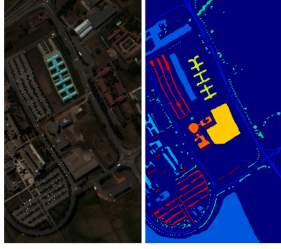
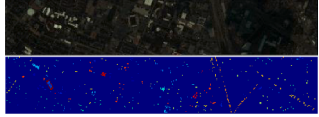
Indian Pines				University of Pavia				Houston			
											
(a)				(b)				(c)			
NO.	Color	Class	Training Total	NO.	Color	Class	Training Total	NO.	Color	Class	Training Total
1		Alfalfa	5 46	1		Asphalt	5 6631	1		Healthy grass	5 11251
2		Corn-notill	5 1428	2		Meadows	5 18649	2		Stressed grass	5 1254
3		Corn-mintill	5 830	3		Gravel	5 2099	3		Synthetic grass	5 697
4		Corn	5 237	4		Trees	5 3064	4		Tree	5 1244
5		Grass-pasture	5 483	5		Painted metal sheets	5 1345	5		Soil	5 1242
6		Grass-trees	5 730	6		Bare Soil	5 5029	6		Water	5 325
7		Grass-pasture-mowed	5 28	7		Bitumen	5 1330	7		Residential	5 1268
8		Hay-windrowed	5 478	8		Self-Blocking Bricks	5 3682	8		Commercial	5 1244
9		Oats	5 20	9		Shadows	5 947	9		Road	5 1252
10		Soybean-notill	5 972					10		Highway	5 1227
11		Soybean-mintill	5 2455					11		Railway	5 1235
12		Soybean-clean	5 593					12		Parking lot 1	5 1233
13		Wheat	5 205					13		Parking lot 2	5 469
14		Woods	5 1265					14		Tennis court	5 428
15		Buildings-Grass-Trees-Drives	5 386					15		Running track	5 660
16		Stone-Steel-Towers	5 93								
Total			80 10249	Total			45 42776	Total			75 15029
(d)				(e)				(f)			

Fig. 5. False-color maps and ground-truth maps of three HSI datasets. (a) Indian Pines (bands 20, 40, and 60). (b) University of Pavia (bands 47, 27, and 13). (c) Houston (bands 57, 27, and 17). (d)-(f) show the number of training samples.

leaky rectified linear units (Leaky ReLUs). Finally, the Adam optimizer is used to train the network with a learning rate equal to 0.0005 for 600 iterations.

7) MSSGU (Liu et al., 2022): the number of nodes in the second-level codecs is set to 2048. For the subsequent codecs, the numbers of nodes is set to in turn, i.e., [1024, 512, 256]. W_θ maps each node to a 128-D vector to learn the weights of edges. All the activation functions adopt the leaky ReLUs. The network is trained using the Adam optimizer with a learning rate of 0.0005 for 600 epochs.

4.3. Experimental settings

Regarding the hardware system configuration, our experimental environment consists of an Intel(R) Xeon(R) CPU E5-1650 V4 with 3.6 GHz, 64 GB, and an NVIDIA GeForce RTX 2080 Ti graphical processing unit (GPU). Regarding the software system configuration, our SSGU-Net method is implemented using Tensorflow-GPU 1.12 and python 3.5.6 under the Windows10 $\times$ 64 operating system.

In our model, after band selection, the numbers of spectral bands are 100, 80, and 100 for the Indian Pines, University of Pavia, and Houston datasets, respectively. Furthermore, we employ the adaptive momentum optimizer with a learning rate of 0.005. The number of training epochs is set to 3000 to guarantee complete convergence of the proposed model.

Table 5

Sensitivity analysis of different values of parameter m .

m	Indian Pines	University of Pavia	Houston
1	78.54	79.00	61.11
5	83.41	86.15	62.80
10	84.69	82.24	66.35
20	85.56	82.99	68.18
25	84.04	—	74.09
40	—	84.48	—
50	80.53	—	73.37

There are three key parameters to be determined, i.e., the number of spectral bands (m) of each component in the G-ConvLSTM, the proportion (k) of selected samples from the whole set of samples in the graph pooling module, and the number of the network layers (L).

1) *Analysis of the number of spectral bands (m) of each component in the G-ConvLSTM*: We report the experiments for analyzing the influence of different values of m on the performance. Specifically, for the Indian Pines and Houston datasets, m is searched within the set {1, 5, 10, 20, 25, 50} with k fixed at 0.5. For the University of Pavia dataset, m is evaluated in the set {1, 5, 10, 20, 40} (owing to the different numbers

Table 6
Sensitivity analysis of different values of parameter k .

k	Indian Pines	University of Pavia	Houston
0.4	85.47	81.35	72.48
0.5	85.95	82.56	70.19
0.6	86.30	83.80	71.85
0.7	84.13	84.14	75.68
0.8	84.57	88.84	71.65
0.9	84.78	85.41	71.20

Table 7
Sensitivity analysis of different values of parameter L .

L	Indian Pines	University of Pavia	Houston
4	85.89	86.92	75.98
6	86.46	89.57	76.23
8	87.45	87.00	76.46
10	87.21	86.69	74.78

of bands retained after the band selection stage). The OA scores are reported in Table 5, from which it can be seen that the quasi-optimal values of m are 20, 5, and 25 for the Indian Pines, University of Pavia, and Houston datasets.

2) *Analysis of the proportion (k) of selected samples from samples*: The experimental results of the proposed method with different values of k in the set $\{0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$ are shown in Table 6. The optimal value of k are 0.6, 0.8, and 0.7 for the Indian Pines, University of Pavia and Houston datasets, respectively.

3) *Analysis of the number of the network layers (L)*: The network depth, i.e., the number of layers (L), in the proposed SSGU-Net, has a significant influence on the classification performance. We set the L to 4, 6, 8, and 10, respectively, to test the classification performance of the proposed method. The classification accuracies of SSGU-Net are shown in Table 7. The optimal L is 8 for the Indian Pines and Houston datasets, while for the University of Pavia dataset, it is 6. However, the computational complexity and runtime obviously increase with the increase in the network depth. Based on the above analysis, the network depth in the proposed method is fixed as 4. This configuration not only achieves satisfactory classification performance but also enhances the practical applicability of the proposed method.

4.4. Classification performance

Following the parameter settings in Section 4.3, all methods are executed ten times to eliminate any bias caused by random selection of training samples. The mean and the standard deviation of each quantitative metric (i.e., per-class accuracy, OA, AA, and κ) are given in Tables 8–10. In addition, Figs. 6–8 provide a visual comparison of the classification results generated by different methods. A detailed analysis of all methods on three datasets is presented as follows.

1) *Indian Pines*: The experimental results of all methods on the Indian Pines dataset are summarized in Table 8, from which it can be seen that the proposed SSGU-Net model can provide better classification performance than all other considered models. First of all, the G-ConvLSTM can simultaneously learn the spatial relationship between pixels and the correlation between different spectral bands, thus extracting suitable spatial-spectral features for HSI classification. In addition, the semi-supervised graph U-Net is designed by using large amount of unlabeled samples and a small amount of labeled samples to build better classifier for HSI classification. In contrast, for the other methods, GCN and FuNet treat each pixel as a graph node, leading to the loss of spatial-contextual information. Although the MDGCN and DIGCN can dynamically update the graphs and fuse multiscale information for HSI classification, they cannot fully explore the relationship between spectral bands, resulting in the unsatisfactory classification performance. Different from ACGCN,

our method utilizes the graph U-Net as the backbone network, which produces an improvement of 5.22 %, 4.00 %, and 5.79 % in term of OA, AA, and κ , respectively. Besides, it is obvious that the standard deviation of our SSGU-Net method is the smallest among all methods in the OA, AA, and κ values, indicating that its stability is the best.

Fig. 6 shows the corresponding classification maps of all methods on the Indian Pines dataset. It is obvious that the map provided by our SSGU-Net model is the closest to the ground-truth map, as it exhibits fewer misclassification and better class boundaries. The above experimental results demonstrate that the effectiveness of our method is better than that of other methods with the Indian Pines scene.

2) *University of Pavia*: Table 9 reports the experimental results of all classification methods on the University of Pavia dataset. As shown in Table 9, the proposed model achieves the best performance among all methods in terms of OA, AA, and κ on the University of Pavia dataset. More specifically, our method provides 1.58 %, 1.89 %, and 2.21 % improvements when compared with CEGCN.

Besides, by comparing the classification maps of all methods with the ground-truth map in Fig. 7, our method can yield more accurate classification results than other methods for narrow and difficult classes, e.g., classes 7, 8, and 9. This demonstrates the advantage and applicability of our method for HSI dataset in complex urban scenarios.

3) *Houston*: Table 10 reports the classification results of all methods on the Houston dataset. As can be observed, similar to the Indian Pines and University of Pavia datasets, the classification performance of our proposed SSGU-Net method is superior to that of other methods. In particular, we can see that SSGU-Net achieves slightly higher accuracies in the second and sixth classes. Moreover, the OA, AA, and κ of SSGU-Net are 76.26 %, 78.74 %, and 74.37 %, respectively, which is more than 1.5 % compared to the second best classification performance achieved by MDGCN.

The visual performance comparison of all methods on the Houston dataset is presented in Fig. 8. The proposed SSGU-Net achieves the best visual classification result among all methods, which confirms the advantage of our SSGU-Net.

4.5. Discussion of the proposed method and related methods

To further interpret the comparative performance between related works and the proposed method, Fig. 10 shows the comparison of OA, AA, and κ results of the proposed SSGU-Net, MDGCN, and MSSGU for three datasets. The proposed SSGU-Net consistently achieves superior OA, AA, and κ compared to MDGCN on three datasets. This performance advantage is primarily attributed to the semi-supervised graph U-Net framework adopted as the base architecture, which can effectively suppress the influence of noise and enhance classification accuracy. Furthermore, as illustrated in Fig. 10, the proposed method also outperforms MSSGU in terms of OA, AA, and κ . Unlike MSSGU which ignores the spectral correlation in HSIs, the proposed G-ConvLSTM module in our method can effectively capture the joint spatial-spectral features in HSIs, thereby improving classification performance. In summary, the proposed method can both suppress the influence of noise and extract joint spatial-spectral features.

4.6. Sensitivity analysis under different numbers of training samples

To further show the effectiveness of the proposed model, the classification accuracies of all methods under different numbers of training samples are investigated in this subsection. The number of labeled samples per class is set from 5 to 30 at intervals of 5. For classes 7 and 9 in the Indian Pines dataset, where the total sample count is insufficient to meet the training sample requirements, 15 samples are specifically selected. The OA curves obtained by all methods on the Indian Pines, University of Pavia, and Houston datasets are provided in Fig. 9. As expected, the accuracies of all methods are improved with the increase of training samples. Moreover, it can be seen that, for these three HSI

Table 8
Classification results for the Indian pines dataset.

Class	GCN	FuNet	MDGCN	DIGCN	ACGCN	CEGCN	MSSGU	SSGU-Net
1	47.72 ± 23.45	69.92 ± 19.91	98.19 ± 0.99	100.00 ± 0.00	95.39 ± 4.47	91.83 ± 5.68	98.94 ± 1.81	99.73 ± 0.81
2	33.78 ± 12.85	29.02 ± 6.78	55.09 ± 7.47	66.06 ± 13.45	60.94 ± 8.71	54.46 ± 9.30	73.29 ± 16.56	76.79 ± 9.14
3	30.18 ± 13.74	34.16 ± 9.22	94.79 ± 11.70	90.42 ± 15.68	73.24 ± 17.56	52.59 ± 22.38	67.22 ± 14.61	76.28 ± 19.16
4	52.63 ± 17.09	44.32 ± 12.41	84.60 ± 11.11	100.00 ± 0.00	95.98 ± 4.69	82.74 ± 12.54	92.91 ± 7.50	98.48 ± 4.56
5	61.92 ± 7.21	59.00 ± 9.69	76.31 ± 10.41	71.25 ± 4.37	82.87 ± 9.18	68.02 ± 23.93	86.63 ± 6.71	88.08 ± 11.18
6	79.67 ± 8.60	80.32 ± 9.19	87.67 ± 6.32	87.62 ± 19.74	89.78 ± 7.99	96.40 ± 2.31	93.80 ± 4.92	96.51 ± 6.94
7	96.34 ± 3.10	86.96 ± 18.03	100.00 ± 0.00	100.00 ± 0.00	99.03 ± 1.92	96.82 ± 10.55	100.00 ± 0.00	100.00 ± 0.00
8	63.74 ± 31.44	87.60 ± 3.72	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	92.93 ± 8.84	98.03 ± 3.94	100.00 ± 0.00
9	91.48 ± 7.84	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
10	33.99 ± 15.86	49.33 ± 10.67	63.18 ± 12.13	56.72 ± 16.84	73.85 ± 7.65	68.22 ± 13.85	79.09 ± 9.73	80.75 ± 6.35
11	45.78 ± 10.85	49.66 ± 5.24	83.16 ± 2.47	70.57 ± 11.17	76.89 ± 9.12	64.26 ± 17.15	63.51 ± 21.11	76.73 ± 8.55
12	29.53 ± 5.64	27.95 ± 6.88	68.63 ± 11.77	74.24 ± 12.79	83.18 ± 7.18	50.23 ± 16.92	67.23 ± 12.58	92.48 ± 13.22
13	94.32 ± 4.68	96.08 ± 6.19	99.78 ± 0.63	100.00 ± 0.00	96.67 ± 3.84	99.60 ± 1.17	99.85 ± 0.24	100.00 ± 0.00
14	79.72 ± 9.48	74.10 ± 4.36	92.59 ± 9.52	98.90 ± 2.99	93.67 ± 6.63	96.96 ± 2.75	94.07 ± 10.00	100.00 ± 0.00
15	33.72 ± 13.92	30.10 ± 6.49	86.99 ± 17.31	96.67 ± 7.33	96.62 ± 7.50	73.19 ± 16.59	91.10 ± 10.26	94.81 ± 10.30
16	96.16 ± 3.87	92.23 ± 9.84	97.90 ± 1.10	98.43 ± 0.94	95.58 ± 4.31	99.90 ± 0.34	98.52 ± 2.46	97.12 ± 0.81
OA	50.32 ± 1.32	52.31 ± 1.89	80.15 ± 1.70	79.28 ± 3.23	80.67 ± 2.30	71.45 ± 4.16	78.56 ± 4.13	85.89 ± 0.97
AA	60.67 ± 2.49	63.17 ± 1.49	86.80 ± 1.41	88.18 ± 1.58	88.36 ± 1.32	80.51 ± 2.62	87.76 ± 1.69	92.36 ± 1.00
κ	44.54 ± 1.41	46.65 ± 1.92	77.47 ± 1.96	76.54 ± 3.51	78.24 ± 2.50	67.60 ± 4.48	75.80 ± 4.51	84.03 ± 1.10

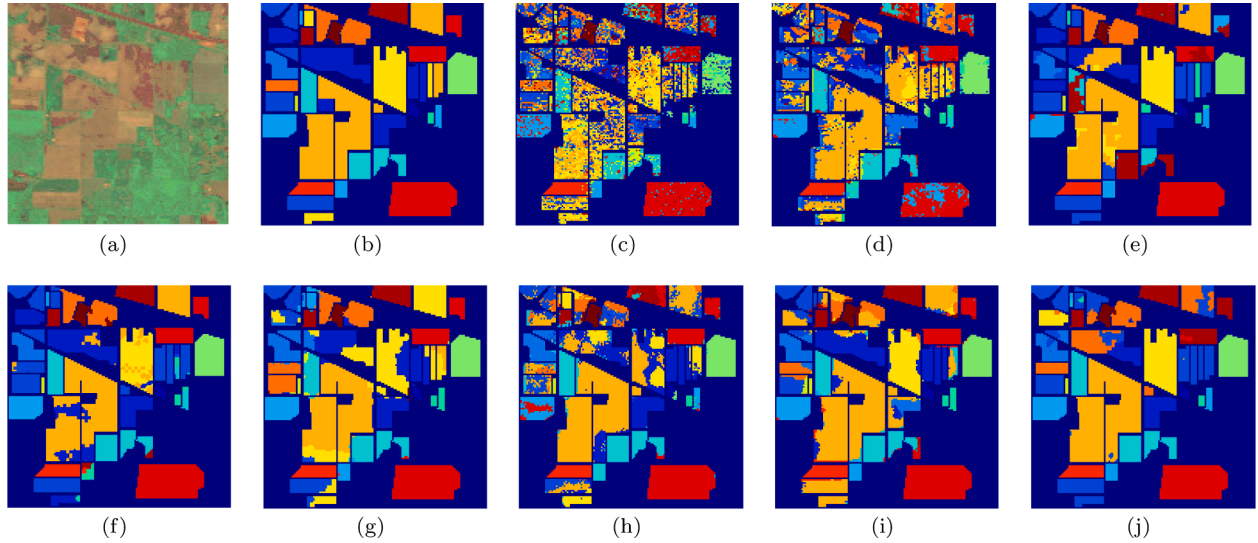


Fig. 6. Classification maps for the Indian Pines dataset. (a) False-color map. (b) Ground-truth map. (c) GCN. (d) FuNet. (e) MDGCN. (f) DIGCN. (g) ACGCN. (h) CEGCN. (i) MSSGU. (j) SSGU-Net.

Table 9
Classification results for the University of Pavia dataset.

Class	GCN	FuNet	MDGCN	DIGCN	ACGCN	CEGCN	MSSGU	SSGU-Net
1	74.08 ± 7.67	63.00 ± 23.36	79.96 ± 4.73	85.53 ± 1.61	83.71 ± 3.87	77.77 ± 9.49	74.82 ± 16.33	77.72 ± 11.20
2	52.28 ± 9.78	78.51 ± 8.83	80.56 ± 3.75	74.78 ± 2.01	75.82 ± 2.57	84.48 ± 4.21	80.76 ± 5.28	84.30 ± 6.23
3	56.65 ± 19.29	62.60 ± 2.40	55.47 ± 3.16	81.26 ± 4.66	83.84 ± 7.48	90.44 ± 9.28	83.93 ± 15.66	73.16 ± 15.60
4	90.09 ± 4.46	63.70 ± 5.63	88.68 ± 2.32	97.61 ± 1.24	97.35 ± 1.41	76.90 ± 5.74	90.29 ± 5.35	87.76 ± 7.60
5	99.21 ± 0.21	99.31 ± 0.71	97.05 ± 3.90	99.93 ± 0.00	99.93 ± 0.35	99.96 ± 0.09	99.98 ± 0.05	100.00 ± 0.00
6	13.84 ± 16.04	55.34 ± 13.32	100.00 ± 0.00	77.05 ± 1.79	82.48 ± 10.66	98.76 ± 1.16	96.70 ± 5.97	99.85 ± 0.31
7	62.67 ± 30.51	78.71 ± 10.50	83.79 ± 7.53	71.50 ± 2.24	78.42 ± 12.27	96.33 ± 4.37	98.93 ± 2.32	99.94 ± 0.13
8	60.71 ± 12.25	64.61 ± 10.88	75.52 ± 2.15	67.36 ± 2.92	73.92 ± 10.73	69.32 ± 16.75	75.58 ± 13.95	86.01 ± 11.55
9	73.99 ± 21.74	99.55 ± 0.23	4.97 ± 0.79	99.97 ± 0.05	99.97 ± 0.05	97.07 ± 4.47	95.62 ± 5.41	99.28 ± 1.50
OA	57.07 ± 5.31	71.47 ± 0.23	80.61 ± 1.53	79.27 ± 0.93	80.97 ± 3.14	84.69 ± 2.31	83.60 ± 1.56	86.27 ± 2.94
AA	64.84 ± 5.16	73.93 ± 1.48	74.00 ± 1.73	83.89 ± 0.58	86.16 ± 4.04	87.89 ± 2.83	88.51 ± 2.49	89.78 ± 2.09
κ	47.08 ± 5.73	62.86 ± 0.56	75.20 ± 1.78	73.55 ± 1.07	75.74 ± 4.03	80.30 ± 2.97	79.05 ± 1.97	82.42 ± 3.60

datasets, the proposed SSGU-Net method consistently yields the highest classification accuracy with different numbers of labeled samples, illustrating the effectiveness of our SSGU-Net method for HSI classification. In particular, compared with other methods, our SSGU-Net method obtains significantly higher accuracy when the number of training samples is less or equal than 15 for the Indian Pines dataset. This demonstrates

that our method is superior to other methods in scenarios dominated by very limited labeled samples. Besides, it can be observed from Fig. 9 that the OA curves rise slowly from 10 to 30 training samples for Indian Pines and University of Pavia datasets. This phenomenon arises because the OAs obtained for the two datasets have reached 95 % with only 10 training samples. All these observations indicate that our SSGU-Net

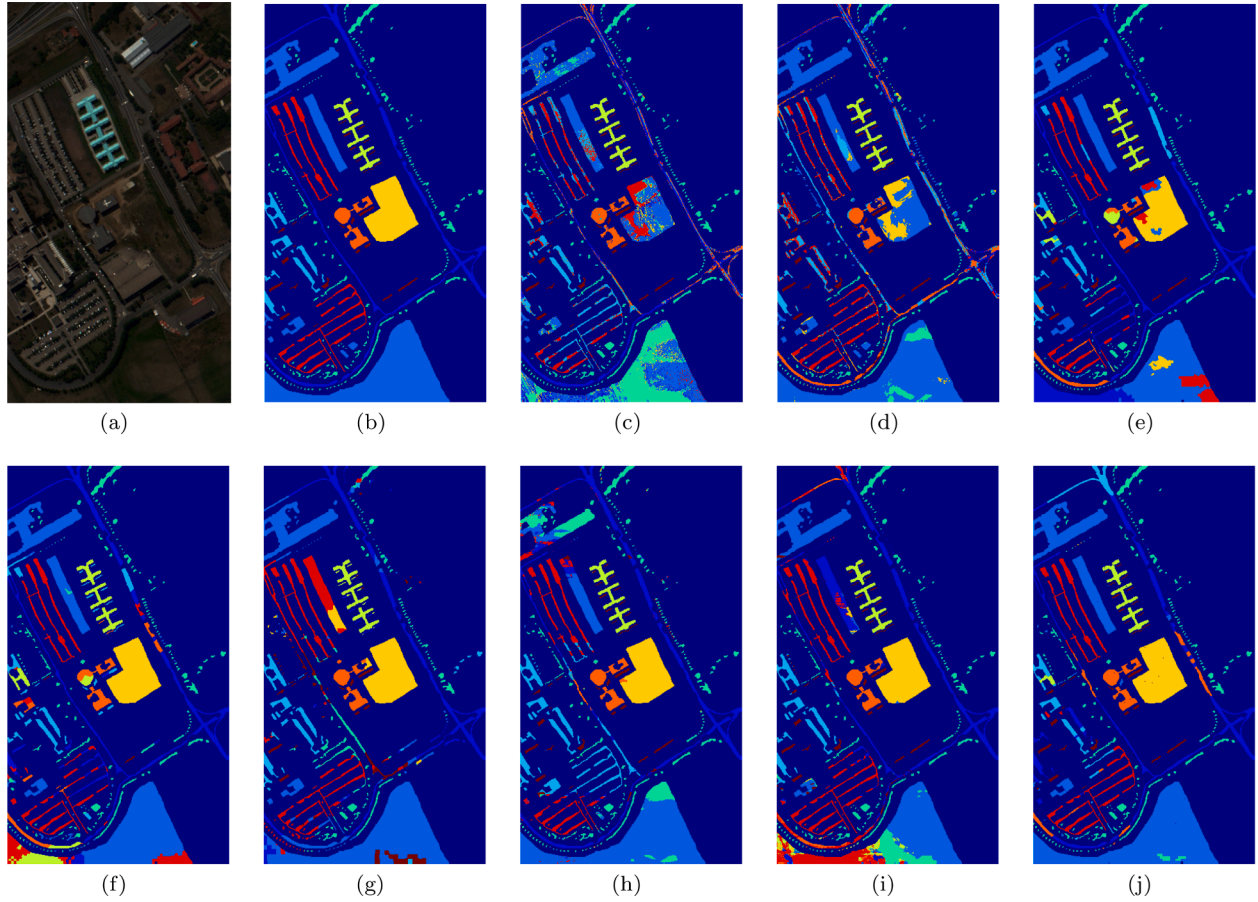


Fig. 7. Classification maps for the University of Pavia dataset. (a) False-color map. (b) Ground-truth map. (c) GCN. (d) FuNet. (e) MDGCN. (f) DIGCN. (g) ACGCN. (h) CEGCN. (i) MSSGU. (j) SSGU-Net.

Table 10

Classification results for the Houston dataset.

Class	GCN	FuNet	MDGCN	DIGCN	ACGCN	CEGCN	MSSGU	SSGU-Net
1	88.17 $\pm$ 8.60	87.08 $\pm$ 6.68	88.11 $\pm$ 6.28	91.49 $\pm$ 6.01	74.77 $\pm$ 9.67	87.17 $\pm$ 3.27	80.21 $\pm$ 6.45	85.71 $\pm$ 6.71
2	83.85 $\pm$ 11.88	88.84 $\pm$ 8.01	89.30 $\pm$ 10.48	72.92 $\pm$ 17.38	80.29 $\pm$ 6.34	63.11 $\pm$ 12.57	42.16 $\pm$ 19.10	100.00 $\pm$ 0.00
3	90.22 $\pm$ 11.04	96.09 $\pm$ 0.72	99.57 $\pm$ 0.40	99.28 $\pm$ 0.00	94.69 $\pm$ 6.85	97.09 $\pm$ 0.68	96.08 $\pm$ 2.23	99.28 $\pm$ 9.96
4	87.32 $\pm$ 4.79	91.38 $\pm$ 2.95	91.72 $\pm$ 2.61	91.53 $\pm$ 1.90	74.41 $\pm$ 5.02	86.29 $\pm$ 5.88	89.87 $\pm$ 2.96	94.35 $\pm$ 2.78
5	91.04 $\pm$ 7.29	90.35 $\pm$ 7.75	98.88 $\pm$ 1.55	94.58 $\pm$ 7.01	95.36 $\pm$ 5.27	94.88 $\pm$ 8.66	94.97 $\pm$ 8.72	96.60 $\pm$ 5.44
6	84.80 $\pm$ 5.26	84.18 $\pm$ 6.95	84.89 $\pm$ 5.72	82.19 $\pm$ 3.78	86.71 $\pm$ 9.41	82.46 $\pm$ 6.10	83.71 $\pm$ 9.71	90.91 $\pm$ 5.63
7	59.88 $\pm$ 18.29	50.76 $\pm$ 13.47	70.36 $\pm$ 15.84	72.30 $\pm$ 9.25	52.95 $\pm$ 13.91	71.54 $\pm$ 20.55	66.18 $\pm$ 3.69	63.10 $\pm$ 7.32
8	36.21 $\pm$ 8.20	39.39 $\pm$ 14.10	34.12 $\pm$ 13.63	39.79 $\pm$ 9.41	61.48 $\pm$ 8.72	45.04 $\pm$ 16.71	31.83 $\pm$ 14.28	42.86 $\pm$ 12.70
9	60.18 $\pm$ 4.84	45.23 $\pm$ 19.50	57.37 $\pm$ 11.21	72.77 $\pm$ 10.40	51.05 $\pm$ 7.68	74.67 $\pm$ 11.17	74.53 $\pm$ 14.59	69.93 $\pm$ 11.40
10	68.42 $\pm$ 14.04	55.69 $\pm$ 19.43	74.81 $\pm$ 9.38	41.10 $\pm$ 12.36	63.49 $\pm$ 6.14	58.67 $\pm$ 11.75	77.23 $\pm$ 19.83	54.66 $\pm$ 10.04
11	69.29 $\pm$ 8.95	47.66 $\pm$ 12.37	64.33 $\pm$ 13.08	60.91 $\pm$ 12.51	58.40 $\pm$ 9.74	75.09 $\pm$ 11.29	82.60 $\pm$ 13.13	79.27 $\pm$ 18.29
12	46.84 $\pm$ 15.06	61.45 $\pm$ 21.03	52.73 $\pm$ 17.77	47.50 $\pm$ 8.38	61.04 $\pm$ 11.03	62.72 $\pm$ 8.38	63.60 $\pm$ 14.44	44.75 $\pm$ 15.49
13	42.91 $\pm$ 7.98	34.72 $\pm$ 7.97	47.33 $\pm$ 14.82	51.16 $\pm$ 18.78	75.27 $\pm$ 8.80	45.17 $\pm$ 31.73	83.50 $\pm$ 8.68	59.91 $\pm$ 23.35
14	82.65 $\pm$ 19.69	86.42 $\pm$ 15.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	97.12 $\pm$ 3.26	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
15	94.72 $\pm$ 1.30	96.39 $\pm$ 0.61	93.91 $\pm$ 8.46	96.66 $\pm$ 6.69	96.28 $\pm$ 2.93	98.65 $\pm$ 0.80	99.33 $\pm$ 1.02	99.69 $\pm$ 0.52
OA	71.13 $\pm$ 1.50	68.54 $\pm$ 2.07	74.69 $\pm$ 1.87	71.86 $\pm$ 1.25	71.30 $\pm$ 4.40	74.45 $\pm$ 3.48	74.27 $\pm$ 2.79	76.26 $\pm$ 2.11
AA	72.43 $\pm$ 1.87	70.38 $\pm$ 1.53	76.49 $\pm$ 1.65	74.28 $\pm$ 0.74	74.89 $\pm$ 4.30	76.17 $\pm$ 4.21	77.72 $\pm$ 2.75	78.74 $\pm$ 2.43
κ	68.83 $\pm$ 1.66	66.02 $\pm$ 2.22	72.68 $\pm$ 2.03	69.62 $\pm$ 1.34	69.05 $\pm$ 4.72	72.36 $\pm$ 3.80	72.19 $\pm$ 3.03	74.37 $\pm$ 2.26

method generally outperforms other methods, verifying its effectiveness and superiority.

4.7. Running times

In this Section, the total sum of the training and testing computational time [seconds (s)] of all methods on three datasets (i.e., Indian Pines, University of Pavia, and Houston datasets) are reported in Table 11. The number of labeled pixels per class is kept identical to

the experiments presented in Section 4.1. Note that GCN, CEGCN, and MSSGU cannot run in the experimental condition in Section 4.3 due to abundant pixels in the University of Pavia and Houston datasets. From Table 11, it is obvious that the GCN method consumes more time than other improved GCN-based methods, owing to the participation of all pixels in training and testing step. CEGCN and MSSGU take the whole HSI rather than small patches as network inputs and all the pixels can be classified in parallel, which results in a shorter classification time. FuNet, as the supervised method, trains the network by a little

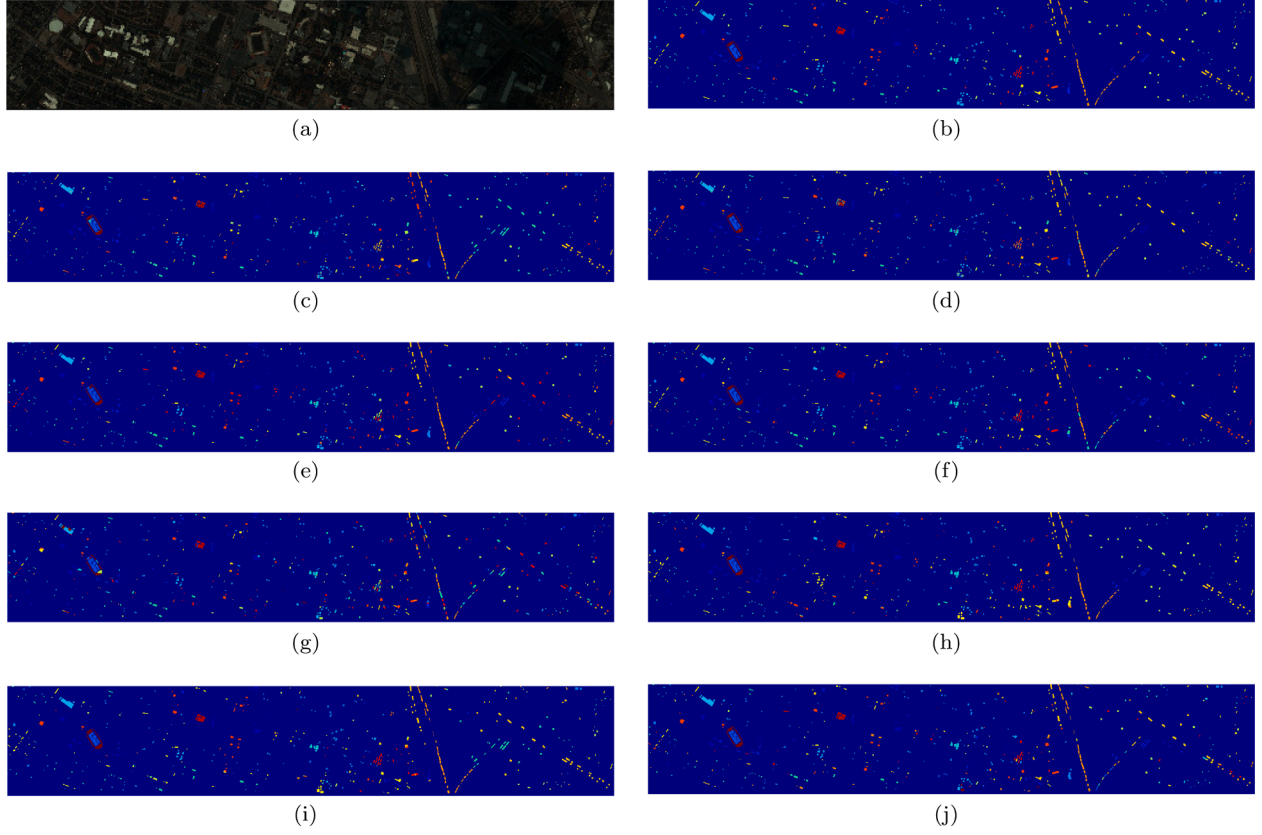


Fig. 8. Classification maps for the Houston dataset. (a) False-color map. (b) Ground-truth map. (c) GCN. (d) FuNet. (e) MDGCN. (f) DIGCN. (g) ACGCN. (h) CEGCN. (i) MSSGU. (j) SSGU-Net.

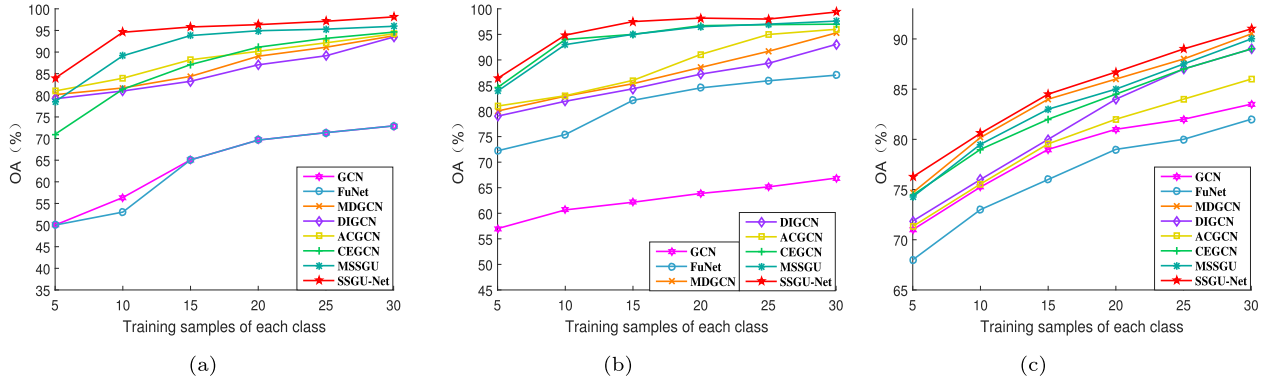


Fig. 9. OA (%) of all methods with different numbers of training samples for three HSI datasets. (a) Indian Pines. (b) University of Pavia. (c) Houston.

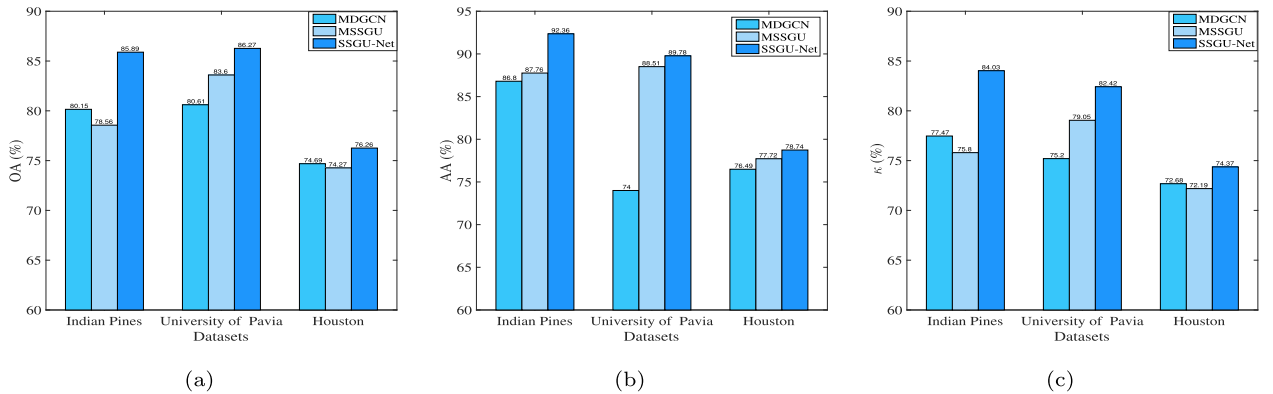


Fig. 10. OA (%), AA (%), and κ (%) for MDGCN, MSSGU, and the proposed SSGU-Net on three datasets. (a) OA. (b) AA. (c) κ .

Table 11
Running time comparison (in seconds) of different methods.

Datasets	GCN	FuNet	MDGCN	DIGCN	ACGCN	CEGCN	MSSGU	SSGU-Net
Indian Pines	318.65	29.13	72.54	260.15	164.70	13.9	27.41	94.21
University of Pavia	–	134.86	255.12	304.79	441.19	67.02	95.61	255.01
Houston	–	29.08	61.91	44.76	193.36	–	–	199.36

Table 12

The OA results of all methods for the original and noisy Indian pines dataset obtained for Gaussian noise with zero mean and various standard deviations σ .

σ	GCN	FuNet	MDGCN	DIGCN	ACGCN	CEGCN	MSSGU	SSGU-Net
–	50.72	53.56	80.08	79.02	81.38	74.32	78.36	88.91
0.001	46.92	52.97	79.12	78.26	80.64	71.48	77.32	87.45
0.01	40.37	43.02	74.52	77.86	77.50	70.30	76.23	86.31
0.1	30.88	19.11	71.69	76.97	77.13	64.65	74.08	84.37
0.3	19.82	13.20	70.23	76.35	76.30	63.71	73.31	83.58
0.5	13.75	11.63	70.12	75.89	75.07	61.48	72.92	82.84
0.7	12.88	11.39	69.26	74.57	74.92	59.27	70.37	82.08

number of labeled samples in the training phase, thus saving time. MDGCN, DIGCN, ACGCN, and SSGU-Net employ SLIC effectively reduce the number of the data and the size of the adjacency matrix, which can reduce the running time. Although more time is needed to run the semisupervised graph U-Net, SSGU-Net achieves the best performance for HSI classification.

4.8. Robustness to Gaussian noise

The goal of this experiment is to evaluate the robustness of all models against Gaussian noise distributions. In this experiment, Gaussian noise with zero mean and various standard deviations (i.e., $\sigma = 0.001, 0.01, 0.1, 0.3, 0.5$, and 0.7) are added to the test HSI data. However, we do not contaminate the training pixels (Nalepa et al., 2021). Table 12 presents the OA results of all methods on the original and noisy Indian Pines dataset. It can be seen that the classification accuracies of all comparison methods decrease with the increase of σ . Moreover, the introduction of higher-intensity noise into hyperspectral test data significantly affects the model's performance. Specifically, MDGCN, DIGCN, ACGCN, and SSGU-Net show better robustness to Gaussian noise compared to GCN, FuNet, CEGCN, and MSSGU because SLIC, as preprocessing operation, can effectively use the spatial context information of HSIs. In addition, SSGU-Net manifests the best robustness against Gaussian noise. The another reason is that the graph pooling module in the semisupervised graph U-Net can reduce interfering unlabeled samples (e.g., noise and outlier) from a set of labeled and unlabeled samples.

4.9. Ablation study

Each novel component in the proposed SSGU-Net method plays a pivotal role. Therefore, a set of ablation studies are conducted to see how they contribute to the overall classification performance. w/o SLIC denotes the proposed SSGU-Net without SLIC. w/o BS refers to the proposed SSGU-Net method without band selection method. A ablation model, called w/o G-ConvLSTM, is obtained by replacing the G-ConvLSTM with the graph convolution operation (Kipf & Welling, 2017). Besides, the SSGU-Net method without the graph U-Net is called w/o graph U-Net. The experimental results are shown in Tables 13 and 14.

1) *Effect of the SLIC and the band selection*: In data preprocessing, the SLIC and band selection are used to reduce the memory requirement and the computational complexity. As observed from Table 13, the proposed SSGU-Net significantly outperforms w/o SLIC by 7.18% OA gains on the Indian Pines while greatly reducing the runtime. In the proposed method, the use of hyperpixels instead of pixels efficiently captures local spatial information to improve classification performance

Table 13

Impact of data preprocessing on the proposed method on three datasets.

Methods	Indian Pines		University of Pavia		Houston	
	OA	Time(s)	OA	Time(s)	OA	Time(s)
w/o SLIC	78.71	2424.84	–	–	–	–
w/o BS	84.75	137.19	82.13	289.41	71.14	238.21
SSGU-Net	85.89	94.21	86.27	255.01	76.26	199.36

Table 14

Ablation results with/without the G-ConvLSTM and the graph U-Net architecture on three datasets.

Methods	Indian Pines		University of Pavia		Houston	
	OA	κ	OA	κ	OA	κ
w/o G-ConvLSTM	83.58	81.47	85.20	81.12	73.99	71.90
	92.33	81.12	89.62	81.12	76.33	71.90
	81.47	81.12	81.12	81.12	71.90	71.90
w/o Graph U-Net	82.56	80.22	84.67	80.03	72.62	70.47
	90.63	80.22	87.40	80.03	74.82	70.47
	80.22	80.03	80.03	80.03	70.47	70.47
SSGU-Net	85.89	84.03	86.27	82.42	76.26	74.37
	92.36	84.03	89.78	82.42	78.73	74.37
	84.03	82.42	82.42	82.42	74.37	74.37

and constructs a small adjacency matrix to reduce computational complexity. In addition, the proposed SSGU-Net improves the OA values by 1.14%, 4.14%, and 5.12%, and reduces the runtime by 42.98s, 34.4s, and 38.85s, respectively, on three datasets compared with w/o BS. This reason is that band selection identifies a subset of the most informative bands by simultaneously discarding redundant or noisy bands, thereby improving classification performance and reducing computational complexity.

2) *Effect of the G-ConvLSTM*: The G-ConvLSTM is designed to jointly learn the correlations between pixels and between spectral bands, thus extracting spatial-spectral features for HSI classification. To demonstrate its effectiveness, we experiment with w/o G-ConvLSTM on three datasets, as shown in Table 14. Compared with w/o G-ConvLSTM, our SSGU-Net method can obtain 2.31%, 1.07%, and 2.27% gains in the OA results for the Indian Pines, University of Pavia, and Houston datasets, respectively, while it also generates 2.56%, 1.30%, and 2.47% improvements in term of κ , respectively. This demonstrates that the spatial-spectral features obtained by the G-ConvLSTM enhance classification performance.

3) *Effect of the graph U-Net architecture*: The graph U-Net architecture (including the graph pooling and unpooling modules) is designed to train better network by using a small amount of labeled samples and

a large number of unlabeled samples. We conduct experiments for analyzing its effect on the classification performance, and the results are reported in Table 14. From the obtained OA scores, we can conclude that the SSGU-Net method with these modules achieves the highest accuracy, with 3.33 %, 1.60 %, and 3.64 % gains for the three datasets compared with w/o graph U-Net, respectively. This demonstrates the effectiveness of graph U-Net in improving the performance of our SSGU-Net model.

5. Conclusion

In this paper, we propose a novel semisupervised graph U-Net with G-ConvLSTM (SSGU-Net) for HSI classification. The proposed method solves the problems of insufficient exploitation of data structure information, limited labeled samples, and high susceptibility to noise and outliers. Specifically, to learn effective spatial-spectral features, the G-ConvLSTM is designed as the feature extractor by simultaneously modeling the geometric structure and the long-term dependence of spectral correlation of HSIs. Furthermore, in contrast to supervised graph U-Nets, the designed semisupervised graph U-Net as the baseline framework fully utilizes both labeled and unlabeled samples to train a network with strong feature extraction abilities. In particular, the graph pooling module retains certain discriminative samples and explores their effective relationship to suppress the effects of noise and outliers. Moreover, the graph unpooling module utilizes the spatial information rather than skip connection to learn the features of the reduced samples, thus restoring the pooled data to its original scale for HSI classification task. Experimental results on three commonly used HSI datasets demonstrate that our SSGU-Net method achieves better performance than other GCN-based HSI classification methods. In future work, we will focus on mitigating data imbalance and generalizing the proposed method to deal with the complex high-order hypergraph.

Data availability

The authors do not have permission to share data.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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